

Mini NLP Project Report

Submitted By: Megha Siju

Reg no: 24UBC141

Class: III BCA A

Course: Natural Language Processing

1. Title

CrisisText: NLP-Based Humanitarian Information Classification for Disaster Response Using TF-IDF and Machine Learning

2. Problem Statement

During natural disasters such as hurricanes, earthquakes, floods, and wildfires, social-media platforms contain a large amount of information. These posts may contain information about affected individuals, infrastructure damage, rescue activities, donations, or general disaster-related events.

Manually analyzing and organizing a large number of social-media posts can be time-consuming. Therefore, this project aims to develop an **NLP-based text classification system** that automatically categorizes disaster-related tweets into different humanitarian information categories.

The developed system takes the text of a tweet as input and predicts one of five categories:

- Affected Individuals
- Infrastructure and Utility Damage
- Not Humanitarian
- Other Relevant Information
- Rescue, Volunteering or Donation Effort

The system is developed as a **prototype decision-support tool** for organizing disaster-related information. It is not intended to replace emergency authorities or guarantee the correctness of emergency information.

3. Objectives

The main objectives of the project are:

1. To identify a socially relevant NLP problem related to disaster response.
2. To use a publicly available disaster-related social-media dataset.
3. To preprocess and clean disaster-related tweet text.
4. To convert textual data into numerical features using TF-IDF.
5. To implement a Multinomial Naive Bayes classifier.
6. To implement a Logistic Regression classifier.
7. To compare the performance of both machine-learning models.
8. To select the better-performing model.
9. To evaluate the final model using accuracy, precision, recall, F1-score, and a confusion matrix.
10. To test the final system using new, unseen text.
11. To identify limitations and suggest future improvements.

The assignment specifically expects students to identify a problem, study suitable techniques, select a dataset and model, implement and test the solution, analyze the results, and identify limitations and improvements.

4. Dataset

4.1 Dataset

The dataset used in this project is **CrisisMMD v2.0**.

CrisisMMD is a publicly available disaster-related social-media dataset containing Twitter posts and associated images collected during major natural disasters.

For this project, only the textual component of the dataset was used because the objective is text-based NLP classification.

4.2 Dataset Source

The dataset was obtained from:

CrisisNLP / Qatar Computing Research Institute (QCRI)

The assignment permits the use of publicly available datasets and requires the source to be identified.

4.3 Dataset Size

The humanitarian dataset was divided into three parts:

Dataset	Number of records used
Training	11,584
Development	2,237
Testing	2,237
Total	16,058

Duplicate tweet IDs were removed from the training data because the same tweet can be associated with multiple images. After duplicate removal, the training data contained **11,584 unique tweets**.

4.4 Attributes

The main attributes in the dataset include:

Attribute	Description
event_name	Name of the disaster event
tweet_id	Identifier of the tweet
image_id	Identifier of the associated image
tweet_text	Text content of the tweet
image	Associated image information
label	Combined text-image annotation

label_text	Text-based humanitarian category
label_image	Image-based category
label_text_image	Combined text-image category

For this project:

Input: tweet_text

Target: label_text

The image-related information was not used because this project focuses on text-only NLP classification.

4.5 Classes / Categories

The original humanitarian task contains eight categories. For this project, related categories were merged into five broader categories:

Final Category	Categories Combined
Affected Individuals	Affected individuals + Injured or dead people + Missing or found people
Infrastructure and Utility Damage	Infrastructure and utility damage + Vehicle damage
Not Humanitarian	Not humanitarian
Other Relevant Information	Other relevant information
Rescue, Volunteering or Donation Effort	Rescue, volunteering or donation effort

Therefore, the final classification problem contains **5 categories**.

4.6 Data Preparation

The following preprocessing steps were performed:

1. Training, development and testing TSV files were loaded using Pandas.
2. The tweet_text and label_text fields were selected.
3. The original eight categories were merged into five categories.
4. Missing values were checked.
5. Tweet text was converted to lowercase.
6. URLs were removed.
7. Retweet indicators such as RT were removed.
8. Twitter mentions such as @username were removed.
9. Hashtag symbols were removed while retaining the hashtag words.
10. Punctuation and unnecessary special characters were removed.
11. Common text encoding problems were handled.
12. Duplicate tweet IDs were removed.
13. The cleaned text was converted into numerical features using TF-IDF.

5. Methodology

The CrisisText project follows a text classification approach using Natural Language Processing and machine learning. The CrisisMMD humanitarian dataset is used, where the tweet text is taken as the input and the humanitarian category is used as the target.

First, the dataset is prepared by merging related original categories into five broader categories: **Affected Individuals, Infrastructure and Utility Damage, Rescue/Volunteering/Donation Effort, Other Relevant Information, and Not Humanitarian**. Duplicate tweets are also removed to improve data quality.

The tweet text is then preprocessed by converting it to lowercase and removing URLs, user mentions, retweet indicators, hashtags, punctuation, and unnecessary characters. This helps reduce noise and prepares the text for feature extraction.

After preprocessing, **TF-IDF (Term Frequency–Inverse Document Frequency)** is used to convert the tweets into numerical features. Unigrams and bigrams are considered, with a maximum of 10,000 features.

Two machine learning algorithms, **Multinomial Naive Bayes and Logistic Regression**, are trained and compared. Naive Bayes achieves **61.6%** development accuracy, while Logistic Regression achieves **65.4%**. Therefore, Logistic Regression is selected as the final model.

The final Logistic Regression model is trained using the combined training and development data and evaluated on the unseen test dataset. Performance is measured using **accuracy, precision, recall, F1-score, and a confusion matrix**. The final model achieves **64.1% test accuracy** and is used to classify new disaster-related tweets into the five humanitarian categories.

6. Implementation

6.1 Programming Language

The project was implemented using:

Python

The assignment requires the application to be implemented using Python.

6.2 Libraries Used

The following Python libraries were used:

Library	Purpose
Pandas	Dataset loading and manipulation
NumPy	Numerical operations
Scikit-learn	TF-IDF, machine-learning models and evaluation
Matplotlib	Graphs and confusion matrix visualization

SciPy	Combining sparse TF-IDF matrices
Regular Expressions (re)	Text cleaning

6.3 NLP Technique

The primary NLP technique used was:

TF-IDF

TF-IDF converts text into numerical vectors based on the importance of words within the documents.

Both single words and two-word combinations were considered using unigrams and bigrams.

6.4 Machine-Learning Models

Multinomial Naive Bayes

Naive Bayes was implemented as a simple baseline model for text classification.

Logistic Regression

Logistic Regression was implemented as the second classifier.

`class_weight="balanced"` was used to reduce the effect of class imbalance.

6.5 Evaluation Methods

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These are among the evaluation measures specifically recommended for classification projects in the assignment.

7. Results

7.1 Class Distribution

After duplicate removal, the training dataset contained **11,584 tweets**.

Category	Number of tweets
Other relevant information	4,266
Not humanitarian	3,299
Rescue/volunteering/donation effort	2,386
Infrastructure and utility damage	921
Affected individuals	712
Total	11,584

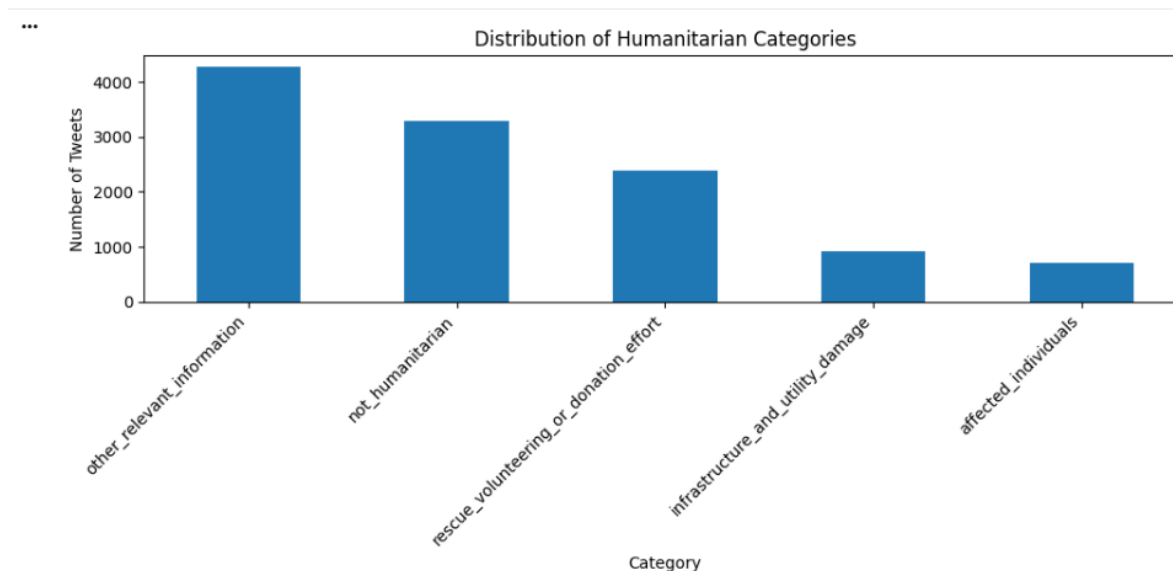


Figure 1: Distribution of Humanitarian Categories

The graph shows that the dataset is not perfectly balanced. *Other relevant information* has the largest number of examples, while *affected individuals* has the smallest number.

7.2 Model Comparison

Model	Development Accuracy
Multinomial Naive Bayes	61.6%
Logistic Regression	65.4%

Logistic Regression achieved **3.8 percentage points higher accuracy** than Multinomial Naive Bayes.

Therefore, Logistic Regression was selected as the final model.

```
print("MODEL COMPARISON")
print("-----")

print("Naive Bayes Accuracy:",
      round(nb_dev_accuracy * 100, 2), "%")

print("Logistic Regression Accuracy:",
      round(lr_dev_accuracy * 100, 2), "%")
```

```
MODEL COMPARISON
-----
Naive Bayes Accuracy: 61.6 %
Logistic Regression Accuracy: 65.4 %
```

7.3 Final Test Results

The selected Logistic Regression model was retrained using the combined training and development data.

Final training size:

13,821 tweets

The model was then evaluated on **2,237 unseen test tweets**.

Final Test Accuracy: 64.1%

Category	Precision	Recall	F1-score
Affected individuals	0.47	0.66	0.55
Infrastructure and utility damage	0.43	0.58	0.50
Not humanitarian	0.65	0.60	0.63
Other relevant information	0.71	0.63	0.67
Rescue/volunteering/donation effort	0.70	0.73	0.72
Macro Average	0.59	0.64	0.61
Weighted Average	0.65	0.64	0.64

7.4 Confusion Matrix

The confusion matrix was used to analyze correct and incorrect predictions.

The diagonal values showed the number of correctly classified test examples:

- Affected individuals → **94**
- Infrastructure and utility damage → **100**
- Not humanitarian → **376**
- Other relevant information → **531**
- Rescue/volunteering/donation effort → **333**

Some notable errors included:

- 127 not humanitarian tweets predicted as other relevant information
- 122 other relevant information tweets predicted as not humanitarian
- 86 other relevant information tweets predicted as infrastructure and utility damage

This indicates that some categories have overlapping language, making them difficult to distinguish using TF-IDF-based classification.

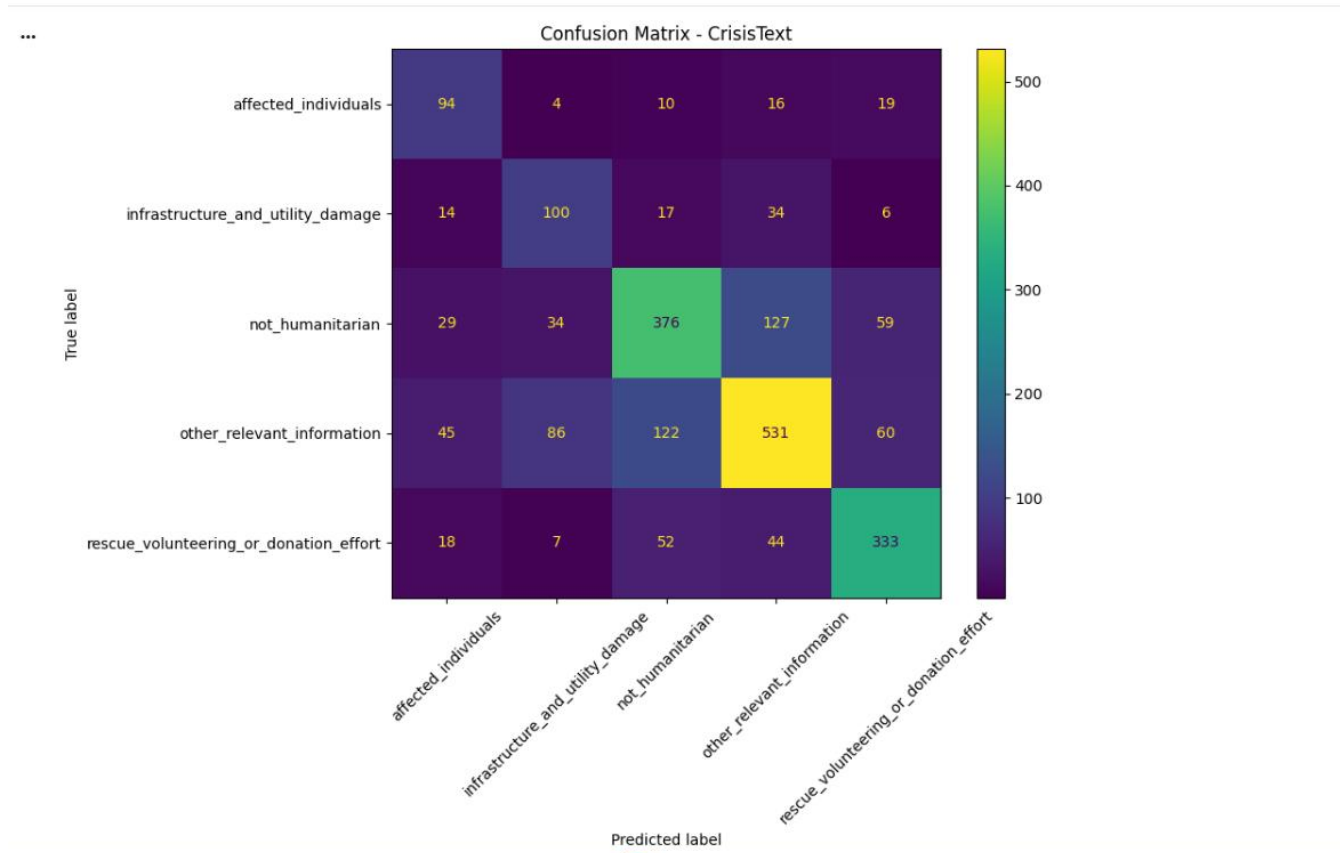


Figure 2: Confusion Matrix of the Final Logistic Regression Model

7.5 Sample Predictions

The final model was also tested with new, unseen sentences.

Input Text	Predicted Category
Many families have lost their homes after the hurricane.	Affected individuals
Volunteers are collecting food and donations for flood victims.	Rescue/volunteering/donation effort
The earthquake damaged roads and several buildings.	Infrastructure and utility damage
Authorities shared information about the ongoing wildfire.	Not humanitarian
I watched a video about the storm on social media.	Other relevant information

These predictions demonstrate how the trained model can classify new text into the five humanitarian categories.

```
*** Tweet: Many families have lost their homes after the hurricane.
Predicted category: affected_individuals
-----
Tweet: Volunteers are collecting food and donations for flood victims.
Predicted category: rescue_volunteering_or_donation_effort
-----
Tweet: The earthquake damaged roads and several buildings.
Predicted category: infrastructure_and_utility_damage
-----
Tweet: Authorities shared information about the ongoing wildfire.
Predicted category: not_humanitarian
-----
Tweet: I watched a video about the storm on social media.
Predicted category: other_relevant_information
-----
```

8. Limitations

The project has the following limitations:

1. **Moderate classification accuracy:** The final test accuracy was 64.1%, so some tweets are incorrectly classified.
2. **Class imbalance:** The number of examples differs considerably between categories.
3. **Limited contextual understanding:** TF-IDF focuses on word importance and does not deeply understand the context or meaning of a sentence.
4. **Overlapping categories:** *Not humanitarian* and *other relevant information* can contain similar vocabulary, resulting in misclassification.
5. **Social-media language:** Tweets may contain abbreviations, informal language, spelling variations, hashtags, and other noisy text.
6. **Ambiguous statements:** Short or context-dependent tweets can be difficult for the model to classify correctly.
7. **Text-only approach:** Although CrisisMMD contains both text and images, this project uses only the textual component.

The assignment also identifies limitations such as limited domains, difficulty with sarcasm, limited vocabulary, incorrect predictions for unseen text, and computational limitations as suitable areas for discussion.

9. Future Scope

The project can be improved in the future through the following approaches:

1. Advanced NLP Models

Transformer-based models such as BERT can be investigated to obtain better contextual understanding.

2. Multimodal Classification

Both tweet text and associated images can be combined to take advantage of the full CrisisMMD dataset.

3. Larger and More Diverse Data

Additional disaster events and larger datasets can be incorporated.

4. Multilingual Support

The system could be extended to process disaster-related information in multiple languages.

5. Real-Time Application

The classifier could potentially be integrated into a real-time disaster-information monitoring system.

6. Web Application

A web interface could be developed where a user enters a disaster-related tweet and receives its predicted category.

7. Better Preprocessing

Advanced tokenization, lemmatization, spelling correction, and domain-specific text processing could be explored.

8. Improved Evaluation

Additional experiments could compare SVM, Random Forest, deep-learning models, and transformer-based approaches.

These directions are consistent with the assignment's suggested future scope, including larger datasets, better preprocessing, advanced models, multilingual support, real-time applications, web/mobile deployment, and better evaluation.

10. Conclusion

The **CrisisText** project developed an NLP-based prototype for classifying disaster-related social-media text into five humanitarian information categories.

The project used the **CrisisMMD v2.0** dataset and performed text preprocessing, duplicate removal, and **TF-IDF feature extraction**. Two machine-learning algorithms, Multinomial Naive Bayes and Logistic Regression, were implemented and compared.

Multinomial Naive Bayes achieved **61.6% development accuracy**, while Logistic Regression achieved **65.4% development accuracy**. Therefore, Logistic Regression was selected as the final model.

After training the selected model using the combined training and development data, the system achieved a **64.1% accuracy** on the unseen test dataset, with a **weighted F1-score of 0.64** and a **macro F1-score of 0.61**.

The project demonstrates that traditional NLP and machine-learning techniques can be used to automatically organize disaster-related social-media information. However, the results also show that some humanitarian categories are difficult to distinguish because of overlapping language. Future work using contextual transformer models, multilingual processing, and multimodal text-image information could potentially improve the system.